

Self-Study DA2: Evaluation of Programs for the Unemployed

Prof. Dr. Johanna Kutz

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Active labour market programs are widely used to improve employment prospects of unemployed, yet estimating their causal effects can be challenging. In this self-study project, you will analyze a hypothetical training program implemented in three regions (A, B, C), each with different assignment and participation mechanisms. In a pilot region, training was offered randomly and participation was voluntary, while in the other regions assignment was determined by caseworkers and participation was mandatory.

The objective of this project is to understand which identification strategies are credible in different settings and how treatment effect estimates change when assumptions are strengthened or violated.

The dataset for this self-study is available on Canvas. Please use the dataset assigned to you. Submit your Python code (main file and functions file) together with a written report in PDF format. The report should not exceed 3 pages in Part I and 3 pages in Part II (tables and figures do not count toward the page limit). The report must clearly present results, interpretation, and a concise explanation of your data preparation and heterogeneity analysis steps.

You are expected to write your own functions for estimating treatment effects, standard errors, test statistics, and for producing plots. For intermediate procedures (e.g., propensity score estimation, matching), standard packages may be used. You may reuse functions from the PC Labs.

Data Description

The dataset contains individual-level observations from a labour market training program. Key variables include:

<code>region</code>	Region A, B, or C
<code>offer</code>	=1 if randomly offered training
<code>takeup</code>	=1 if individual participated in training
<code>outcome</code>	Labour market outcome after five years
<code>age</code>	Age in years
<code>gender</code>	Gender indicator
<code>baseline_income</code>	Pre-program income
<code>unemployment_duration</code>	Duration of previous unemployment

The raw data contains several other covariates, which might be important as well.

Part II: Instrumental Variables, Due: 04.05.2026

In this part, you will use the randomized offer in Region A as an instrument to estimate causal effects and compare these results to Part I, so please drop the other regions.

6. Instrument Validation

Using the `offer` variable:

- Assess whether offers are randomly assigned.
- Estimate the first-stage relationship between offer and take-up.
- Evaluate instrument strength.

Discuss the relevance and exogeneity assumptions.

7. Why not OLS

Discuss: Why would a simple regression of `takeup` (participation) on `outcome` fail to give a credible causal estimate in Region A?

8. IV Estimation

Estimate:

- ITT (effect of being offered training),
- LATE using the Wald estimator,
- 2SLS estimates controlling for covariates.

Interpret each estimand carefully, report standard errors for all estimands.

9. Comparing Identification Strategies

Compare Estimates from Part I and IV/LATE estimates from Part II.

- Which identification strategies are valid and where?
- How do ATE, ITT, and LATE differ, and what populations do they refer to?
- What are the policy implications?